Tree-Based Machine Learning Estimation of Near-Real-Time Diffuse Solar Irradiance From Himawari-8 Satellite Data

Yunhui Tan, Quan Wang, and Zhaoyang Zhang

***Abstract*— Diffuse solar radiation (R_d), one of the two components of solar radiance, has been shown to have a fertilizing effect on plants and, thus, has implications for terrestrial carbon sinks and the water cycle. Currently available R_d data are mostly at hourly or daily scales, but their production very often depends on additional meteorological data. Furthermore, very few works have estimated R_d in near-real time, as dynamically changing clouds and aerosols pose great challenges for instantaneous estimation. In this study, we adopt the Himawari-8 satellite data as the main input, to estimate the near-real-time R_d using four tree-based machine learning algorithms, namely gradient boosting regression tree (GBRT), random forest (RF), XGBoost, and LightGBM, similarly. Although all models performed well, with R^2 around 0.80 and rRMSE at 32.97%–37.2% for random tenfold cross-validation (CV), LightGBM has the highest efficiency. Further ground validation on independent datasets showed that the results are highly consistent across different time scales. Additional analyses revealed that different models behave differently under clear and nonclear sky conditions, but all perform poorly under clear sky conditions. Taken together, we foresee that this efficient method, based on Himawari-8 geostationary satellite data, can be widely applied to estimate near-real-time R_d in the future.**

***Index Terms*— Diffuse solar radiation, gradient boosting regression tree (GBRT), Himawari-8 satellite, LightGBM, random forest (RF), XGBoost.**

I. Introduction

GLOBAL solar radiation (R_s), as the primary source of energy on the Earth, is critical for various applications such as energy budget balancing, agricultural meteorology, and climate monitoring [1], [2], [3], [4], [5]. It consists of diffuse solar radiation (R_d) and direct solar radiation. Compared with direct radiation, R_d is more efficient in enhancing plant physiological processes [6], [7], which consequently plays a key role in land carbon assimilation and the water cycle [8], [9]. As a consequence, it is essential to ensure accurate estimation of R_d.

To date, the most reliable way to provide the true value of R_d is from ground observations, and there are several ground radiance observation networks available as data sources, such as the baseline surface radiation network (BSRN) [10], the surface radiation budget network (SURFRAD) [11], the World Radiation Data Center (WRDC) [12], and the southern African universities radiometric network (SAURAN) [13]. Unfortunately, the R_d is still relatively rare in the ground observations [12], with many sites having only the R_s but lacking the R_d components due to the lack of expensive equipment or the lack of instrument maintenance resulting in the data no longer being updated. Furthermore, relying on sites alone poses challenges when applied to large spatial scales [14]. Alternatively, there are some radiation products or reanalyzed data, such as the clouds and Earth's radiant energy system (CERES), which provides the R_d with a resolution of $1°h^{-1}$ [15]; and fifth-generation ECWMF reanalysis (ERA5) supplies the R_d of $0.25°h^{-1}$ [16]; while the modern-era retrospective analysis for research and applications, Version 2 (MERRA-2) provides the radiance of $0.5° \times 0.625°$ h^{-1} [17]. However, their temporal resolution is hours or even days, and their spatial resolution is very coarse as well. Taken together, there exists considerable potential for advancing the development of real-time, large-scale spatial R_d products.

As a recent advance, remote-sensing-based methods can overcome the aforementioned limitations and provide R_d with relatively high spatiotemporal resolution, mainly including the following categories: first, the separation of R_s into R_d and direct solar radiance. There are many empirical methods for estimating the ratio between R_d and R_s [18], [19], and then calculating the R_d according to the ratio. However, this method is usually employed when R_s is already known, and these empirical formulas may vary between different locations [20]. Second, R_d is estimated directly by the physical or simplified radiative transfer model or based on the offline lookup table generated by the radiative transfer simulation. For example, Leng et al. [21] used the lookup table method to estimate the components of R_s. However, this approach requires a lot of atmospheric-related auxiliary data. Third, by the popular machine learning methods. To name a few, Lou et al. [22] predicted the R_d from meteorological variables through a boosted regression tree (BRT), while Hao et al. [23] estimated the R_s and photosynthetically active radiation (PAR) and its diffuse component through the random forest (RF) from the DSCOVER/EPIC observation. Furthermore, Lu et al. [24] combined the radiative transfer model with machine learning, including RF, eXtreme Gradient boosting (XGBoost), deep neural networks (DNNs), and convolutional neural networks (CNNs), to estimate the R_d.

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R_d refers to scattered radiation, and clouds and aerosols in the atmosphere are usually the main contributors to the R_d in the environment [25], [26]. The Himawari-8 satellite, which was launched by the Japan Meteorological Agency (JMA) in October 2014, could provide near-real-time observations over East Asia and Australian regions, offering privileged monitoring for real-time weather conditions [27]. Currently, the official Himawari-8 radiance products provide the R_s and PAR at intervals of 10-min (Level 2) and 1-h (Level 3) intervals according to the algorithm of Frouin and Murakami [28], but there is no R_d component. Likewise, Yu et al. [29] adopted the look-up-table-based method to estimate the R_s and direct solar radiance from Himawari-8 spectral data. In addition, Li et al. [30] used the improved radiative transfer method to estimate the R_s from the Himawari-8 satellite, while Hou et al. [31] estimated the R_s from Himawari-8 observations using RF. Furthermore, Peng et al. [32] used the artificial neural network (ANN) to estimate the R_s from the Himawari-8 satellite, whereas Ma et al. [33] combined the ANN and radiative transfer model to estimate the R_s. Overall, these machine learning methods for estimating radiation from the Himawari-8 satellite pay little attention to the R_d component, albeit Letu et al. [34] provided the R_d component by combining the radiative transfer model and machine learning methods, which involves complex radiative transfer calculations and requires the construction of a simulated radiation database that consumes a lot of memory. To date, only a few studies have been carried out using separate computational methods for estimating the diffuse radiation from the Himawari-8 satellite.

Machine learning methods have their advantages and disadvantages [35]. The tree-based machine learning methods are widely used because the predictive ability is better than deep learning on typical tabular data [36]. Hassan et al. [37] explored the tree-based ensemble methods in solar radiation predictions based on meteorological variables such as sunshine hours and clearness index. However, they are rarely used to estimate diffuse radiation based on satellite data. Therefore, the objective of this study is to utilize the four tree-based machine learning methods, including gradient boosting regression tree (GBRT), light gradient boosting machine (LightGBM), XGBoost, and RF, to estimate the R_d directly from the Himawari-8 satellite data. This article is organized as follows: the ground observations, satellite data, and the methods used in this study are introduced in Section II, and the results and the discussion are presented in Sections III and IV, respectively. Finally, a summary is presented in Section V.

II. Materials and Methodology

A. Main Datasets

The Himawari-8 new-generation Japanese geostationary satellite launched on October 7, 2014, can provide high-frequency monitoring, the full-disk observes the East Asia and Australian regions at 10-min, and the observations can even reach 2.5 min over the Japan area since July 7, 2015 [38]. The official L1 grid product provides the reflectance at the top-of-atmosphere and geolocation information in NetCDF format files with a spatial resolution of 5 × 5 km. The study utilized this dataset for the years 2018 and 2019.

TABLE I
Information of All Sites Used in This Study

Code	Latitude (°)	Longitude (°)	Elevation (m)	Source
ASP	-23.80	133.89	547	BSRN
DWN	-12.42	130.89	32	BSRN
FUA	33.58	130.38	3	BSRN
LAU	-45.05	169.69	350	BSRN
NEW	-32.88	151.73	18.5	BSRN
SAP	43.06	141.33	17	BSRN
TAT	36.06	140.13	25	BSRN
YUS	23.49	120.96	3858	BSRN
SL023034	-34.95	138.52	2	BOM
SL015590	-23.80	133.89	546	BOM
SL003003	-17.95	122.23	7.4	BOM
SL091148	-40.68	144.69	95	BOM
SL014015	-12.42	130.89	30.4	BOM
SL008315	-28.80	114.70	29.7	BOM
SL012038	-30.78	121.45	365.3	BOM
SL086282	-37.66	144.83	113.4	BOM
SL039083	-23.36	150.48	10.4	BOM
SL032040	-19.25	146.77	4.3	BOM
SL072150	-35.16	147.46	212	BOM

The surface albedo was obtained from the MCD43A3 product, which provides a daily L3 albedo product with a spatial resolution of 500 m. This product provides both directional hemispheric reflectance (Black Sky Albedo, BSA) and bihemispheric reflectance (White Sky Albedo, WSA) for each of the MODIS surface albedo bands (band 1 through band 7) as well as three broadband spectral bands (Visible, Near-infrared, and Shortwave) [39]. To be consistent with the spatial resolution of the Himawari-8 satellite data, we resampled them to 5 km in the Google Earth Engine (GEE) platform.

The ground radiation observations from the Bureau of Meteorology (BOM) in Australia and BSRN are used in this study. There are a total of 19 sites, eight sites from the BSRN and 11 sites from BOM, the information is shown in Table I. All sites had total, direct, and diffuse solar radiation at 1-min intervals. The raw 1-min data were quality-checked according to the method of Yang and Gueymard [40]. Finally, they were averaged to 10-min intervals for matching with Himawari-8 satellite data.

B. Data Preprocessing

Eventually, there were 34 features from the MODIS and Himawari-8 satellite data. After quality checks and data matching, there were a total of 369 323 and 368 208 samples in the year 2018 and 2019, respectively. The min-max normalization method was used for data normalization, which avoids the impact of too large differences between feature values on the robustness of the model. Finally, the correlation between

each feature and the R_d was calculated, and the variables that have significant positive or negative effects on R_d ($p < 0.01$) were selected. However, the collinearity of the independent variables must be taken into account to avoid duplication of the information conveyed by the independent variables. Therefore, the variance inflation factor (VIF) method was selected to diagnose collinearity among the selected variables, variables are finally selected when VIF < 10 [41]. The final features included in the model had a total of eight variables, which were albedo_01 (reflectance of blue band), satellite zenith angle (SAZ), satellite azimuth angle (SAA), solar zenith angle (SOZ), solar azimuth angle (SOA), Albedo_BSA_Band3, Albedo_WSA_Band2, Albedo_WSA_Band7.

C. Machine Learning Models

GBRT, LightGBM, RF, and XGBoost are four common tree-based machine learning methods used in this study. The GBRT is a type of method that is based on the decision tree through iterative optimization, it could automatically find the nonlinear interactions through the decision trees and it has better predictive ability than a single decision tree [42], [43]. RF model is an effective ensemble machine learning method based on decision trees [44], it is also operated by constructing a mount of regression trees [45]. Compared to RF, XGBoost is a relatively new implementation of the gradient-boosting decision trees, that simultaneously optimizes the loss function while constructing the additive model. That means, XGBoost supports column sampling and parallel processing, which significantly improves the efficiency of the algorithm [46]. Compared with XGBoost, the LightGBM uses a histogram-based algorithm that divides continuous eigenvalues into n intervals and selects the dividing points in the n intervals to reduce the resources consumed by storage to achieve improved accuracy [47], the details of the algorithm could be referred to Zhou et al. [48].

All of these four methods use tree-based learners and can be used for both classification and regression problems. However, their architecture and algorithms are fundamentally different, resulting in differences in performance and accuracy. In this study, all methods are implemented on Python 3.8 using the Scikit-learn toolbox, and the hyperparameters are tuned by using the Bayesian optimization method [49]. The model hyperparameters setting of four models and the final parameters of the models are listed in Table II.

D. Evaluation Strategy

To evaluate the accuracy of the models comprehensively and prevent overfitting, different cross-validation (CV) methods are used [50]. In this study, we used random tenfold CV, time-based CV, and site-based CV in the 2018 dataset, respectively. In the random tenfold CV, the data in the year 2018 was randomly divided into ten parts, one part as the test set and the other part as the training set. For the time-based CV, the data were split into ten parts according to the Julian day of the year in 2018, and one part was used for testing. For the site CV, we used the leave-one-station-out method, which means that the data from one station were used for testing, and data from other sites were used for training.

TABLE II
IMPORTANT MODEL PARAMETERS SETTING

Name	Parameter	Range	Final Value
RF	n_estimators	(2,500)	420
	max_depth	(2,25)	25
	min_samples_split	(2,20)	10
	min_samples_leaf	(10,20)	18
XGBoost	n_estimators	(2,500)	354
	max_depth	(2,25)	14
	learning_rate	(0.0001,0.1)	0.03
	Colsample_bytree	(0.5,1)	0.82
	reg_alpha	(1e-5,10)	0.226
LightGBM	n_estimators	(2,500)	317
	max_depth	(2,25)	21
	learning_rate	(0.0001,0.1)	0.1
	num_leaves	(50,500)	394
GBRT	n_estimators	(2,500)	446
	max_depth	(2,25)	12
	learning_rate	(0.0001,0.1)	0.1
	min_samples_leaf	(10,20)	10
	min_samples_split	(2,20)	15

At the same time, we used the 2019 dataset as the independent dataset for testing the robustness of the four models and evaluated them at different time scales, including seasonal and daily scales. Besides, to investigate the performance of the models under different weather conditions. In this study, we classified the dataset into five different categories of weather conditions, namely, Clear, I-Clear, Intermediate, I-Overcast, and Overcast according to the sky index (Si) referenced in Tan et al. [46]. In addition, the clearness index (Kt) is defined as the ratio of the global solar irradiance measured at ground level to its counterpart estimated at the top of the atmosphere [51]. It is often used as a parameter in empirical models to estimate R_d, and it is also used to determine different weather conditions. The higher the Kt value, the fewer clouds, which means that the direct component dominates [52]. Therefore, we investigated the model performance under different Kt values.

This study used five statistical factors, namely, the determination coefficient (R^2), root mean square error (RMSE), relative RMSE (rRMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE) to evaluate the accuracy of the R_d estimation on the additional testing dataset. The formulas are as follows:

$$R^2 = \frac{[\sum_{i=1}^{n}(y_{\text{obs}i} - y_{\text{obs_mean}}) \times (y_{\text{est}i} - y_{\text{est_mean}})]^2}{\sum_{i=1}^{n}(y_{\text{obs}i} - y_{\text{obs_mean}})^2 \times \sum_{i=1}^{n}(y_{\text{est}i} - y_{\text{est_mean}})^2} \tag{1}$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(y_{\text{obs}i} - y_{\text{est}i})^2}{n}} \tag{2}$$

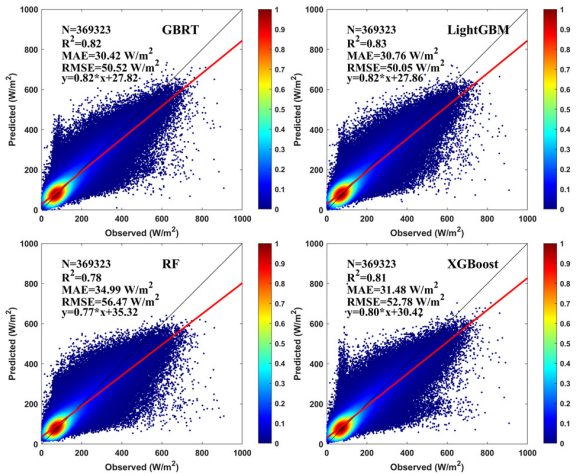


Fig. 1. Random tenfold CV results on a dataset of the year 2018 for four methods: GBRT, LightGBM, RF, and XGBoost.

$$r\mathrm{RMSE}(\%) = 100 \times \mathrm{RMSE}/y_{\mathrm{obs_mean}} \tag{3}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_{\mathrm{obs}i} - y_{\mathrm{est}i}| \tag{4}$$

$$\mathrm{MAPE}(\%) = \frac{100}{n}\sum_{i=1}^{n} |(y_{\mathrm{obs}i} - y_{\mathrm{est}i})/y_{\mathrm{obs}i}| \tag{5}$$

where the parameters $y_{\mathrm{obs}i}$ and $y_{\mathrm{est}i}$ represent the ground-measured and estimated R_d, whereas $y_{\mathrm{obs_mean}}$ and $y_{\mathrm{est_mean}}$ are the respective mean values, and n is the total number of samples in the dataset.

III. Results

A. Model Performance Under Different CV Methods

We compared the performance of the four machine learning methods, and the results of the random tenfold CV on the 2018 dataset are shown in Fig. 1. The x-axis standards for the observed R_d on the ground platform, and the y-axis represents the predicted R_d by GBRT, LightGBM, RF, and XGBoost, respectively. In general, four methods can well predict the R_d under the random tenfold CV, with R^2 around 0.8, and RMSE around 50 W/m^2. Among them, LightGBM has the smallest MAE of 30.76 W/m^2 and RMSE of 50.05 W/m^2.

At the same time, the leave-one-station-out CV and the time-based CV results are summarized in Table III. For the leave-one-station-out CV, four methods have similar performance as well, but the RF has the lowest RMSE of 78.02 W/m^2 and MAE of 51.44 W/m^2. For time-based CV, the RF again has the best performance. In comparison, all four methods have worse performance for leave-one-station-out CV and time-based CV. In addition, we counted the running time of four models during the whole CV process and all models are executed on the same computer, we found that LightGBM takes the shortest time, and is about 23× faster than GBRT, which takes the longest time.

B. Model Performance on the Independent Dataset and on Different Time Scales

In addition to using different CV methods for evaluating the models, we used an additional dataset in 2019 as the independent test dataset to evaluate the accuracy of the models and whether they can correctly catch the dynamics in R_d over different time scales. As shown in Fig. 2, the four machine learning methods have similar performance, among which RF has the best performance with R^2 of 0.7, RMSE, and MAE are 68.23 and 43 W/m^2, respectively. In summary, the four methods can predict R_d well on the independent test dataset, but there will be phenomena of underestimation at high R_d values and overestimation at low R_d values for all models.

Furthermore, we analyzed the model performance on this independent dataset separately for seasonal and daily scales. From Fig. 3, we can see that the four models predict R_d well under different seasons. If the place is in the northern half of the Earth, it has spring in March, April, and May, summer in June, July, and August, autumn in September, October, and November, and winter in December, January, and February. If it is in the southern half, then it is the other way around. The prediction of the RF model is slightly better than the other three methods in all seasons. Similarly, the four methods can track the daily dynamic variation very well for each site as shown in Fig. 4.

TABLE III
Statistical Results for Different CV Methods

Methods	Time (h)	CV types	R^2	RMSE (W/m^2)	MAE (W/m^2)	rRMSE (%)	MAPE (%)
GBRT	42.86	Random	0.82	50.52	30.42	33.27	24.47
		Site-based	0.54	83.12	55.30	54.75	51.78
		Time-based	0.68	67.82	44.75	44.67	38.39
LightGBM	1.86	Random	0.83	50.05	30.76	32.97	25.19
		Site-based	0.55	81.49	53.52	53.67	48.66
		Time-based	0.68	67.74	44.81	44.62	38.89
RF	22.66	Random	0.78	56.47	34.99	37.20	28.66
		Site-based	0.58	78.02	51.44	51.39	45.97
		Time-based	0.70	65.50	42.35	43.14	35.31
XGBoost	13.58	Random	0.81	52.78	31.48	34.77	25.78
		Site-based	0.55	80.75	53.64	53.19	49.30
		Time-based	0.69	66.46	43.68	43.77	37.19

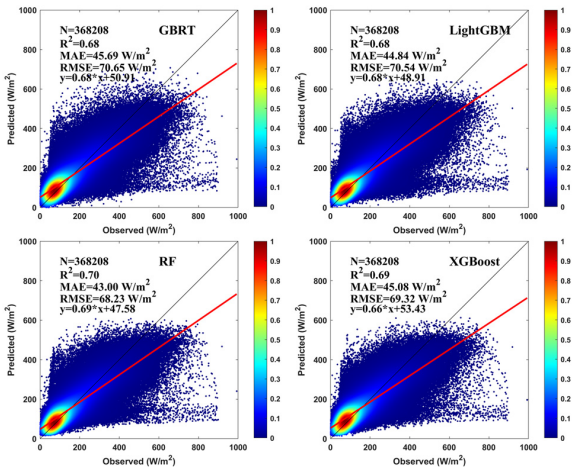


Fig. 2. Same as Fig. 1, but for the independent dataset of the year 2019.

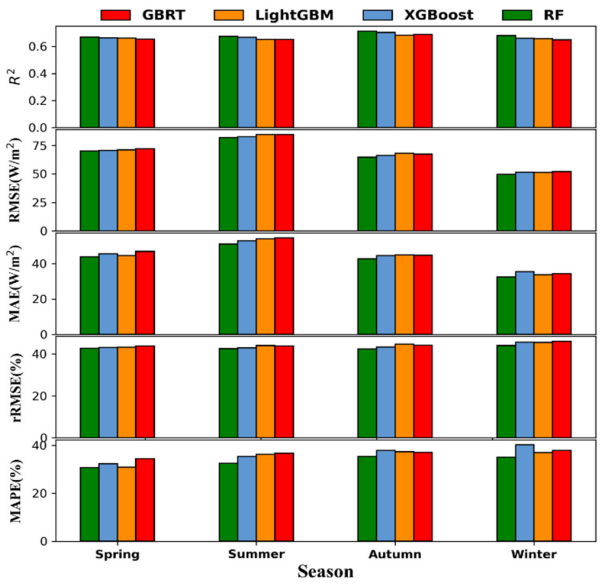


Fig. 3. Relative statistical results on an independent dataset of the year 2019 for four seasons.

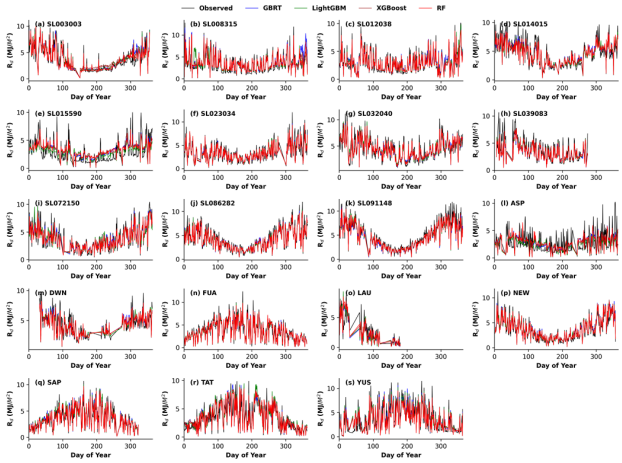


Fig. 4. Model performance on a daily scale on an independent dataset of the year 2019 for each site.

C. Model Performance Under Different Weather Conditions

The model performance on the independent dataset for five weather conditions is illustrated in Fig. 5 (left). All models have the worst performance under the Clear type and the best performance under the Intermediate type with the smallest rRMSE and MAPE. These four methods perform consistently under the same weather conditions.

In addition to evaluating the performance of the original four models under different weather conditions on the independent dataset of the year 2019, we also evaluated the separate models established by the data under clear sky and nonclear sky conditions, where we defined all the categories except the clear-sky category into the nonclear-sky category here. As depicted in Fig. 5 (right), the solid histograms represent the performance of the original four models, while the dashed histograms represent that of separate models. The original four models have similar overall performance, with all models performing better under nonclear-sky conditions as indicated by their lower rRMSE and MAPE. In addition, by training the models considering different weather conditions, we found

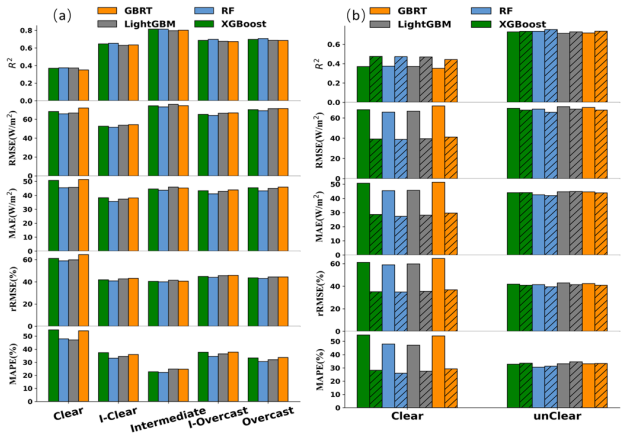


Fig. 5. Model performance under different weather conditions on the independent dataset. (a) Comparison between four original models, all without clear/nonclear separation. (b) Comparison between original models (solid histograms) and models established by the data under clear/nonclear separation (dashed histograms).

that the accuracies of the models can be improved greatly under clear-sky, while the improvement is not obvious under nonclear-sky.

IV. Discussion

A. Overall Performance of the Predicted Models

In this study, we utilized four tree-based machine learning methods, namely GBRT, LightGBM, RF, and XGBoost, to estimate the diffuse solar radiation and also conducted a comprehensive evaluation. In general, the four models have similar performance regardless of time-based, site-based, or independent test data. However, it is worth noting that, the LightGBM model has the shortest run time, followed by XGBoost, then RF, and finally GBRT. This is because LightGBM is an optimization model of XGBoost, which has improved information gain, decision tree construction, and feature parallelism [53], [54]. This is similar to the study by Zhou et al. [48], which suggests that LightGBM has higher computational efficiency compared to XGBoost for solar radiation estimation. Compared to RF, XGBoost has achieved an almost 50% improvement in efficiency in our study, owing to the parallel process during modeling [55]. We also conducted the temporal and spatial CV to evaluate the reliability of the four models and found that the spatial CV performed the worst for all four models, which is consistent with many studies and is a problem for machine learning models that depend on the training datasets, and transfer learning can be one way to solve this problem [56].

To date, relatively few R_d products have been offered globally or in East Asia. The diffuse radiation products we currently know include CERES with a resolution of 1° × 1°, MERRA-2 with 0.5° × 0.625°, and ERA5 with 0.25° × 0.25° [12]. A comparison of the ERA5 and JiEA R_d products using 39 stations from the WRDC, and the China Meteorological Administration (CMA) revealed that ERA5 significantly underestimates R_d. As a further evaluation, we compared a case of R_d estimation by our method with ERA5 due to its higher spatial resolution at the TAT site in Japan [See Fig. 11 below for location]. The ERA5 R_d values were

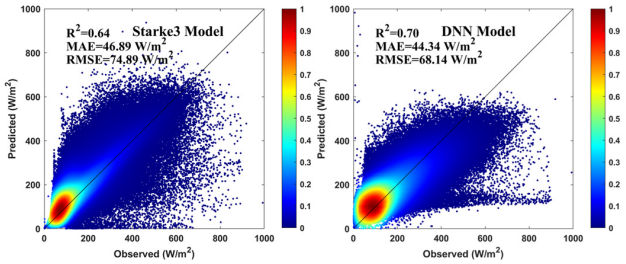


Fig. 6. Results of Starke3 and DNN models on the independent dataset in the year 2019.

particularly underestimated in the morning of August 1, 2019, but overestimated in the afternoon. In comparison, the predicted R_d from the four machine learning methods were in good agreement with the ground observations. Consequently, these four methods can be used to estimate the R_d at the site scale, especially for sites where R_d data are missing due to lack of maintenance. While these four methods are highly reliable, we recommend using the LightGBM model for the operational efficiency of the program.

B. Comparison Between the Empirical Model and DNN Model

To comprehensively evaluate the performance of the tree-based models, we compared them with the empirical Starke3 model and the simple DNN model. The Starke3 model is one of the most accurate empirical models for estimating the R_d [18]. A simple DNN with one input layer, one hidden layer, and one output layer was used for this comparison. Their respective performance on an independent dataset in the year 2019 is shown in Fig. 6. Compared with the predicted results in Fig. 2, the results of the empirical Starke3 model are worse on the independent dataset, while the DNN model has a similar performance to the RF model, but with a slightly higher MAE. Therefore, we can conclude that tree-based machine learning methods are more accurate than the empirical Starke3 model and have similar performance to the DNN model.

C. Comparison With the ERA5 Product

To further evaluate the accuracy of the tree-based machine learning approaches, we compared their estimates with the hourly ERA5 analysis of solar diffuse component products on an independent dataset in 2019. After matching, there are a total of 62 268 samples. The results are shown in Fig. 7, the R^2, MAE, and RMSE are 0.3, 69.05 W/m^2, and 104.57 W/m^2, respectively. These results are much worse than all the tree-based methods in this study. Similarly, RF performs best among all four tree-based models, with the smallest statistical error.

D. Model Bias Analyses: Feature Importance and Feature Dependence

To further determine the error sources of the models, we analyzed the bias for four models. The bias variation with R_d is shown in Fig. 8. We noticed that the bias trend for the four models was similar, all of them were changed from positive to negative, indicating that all models have

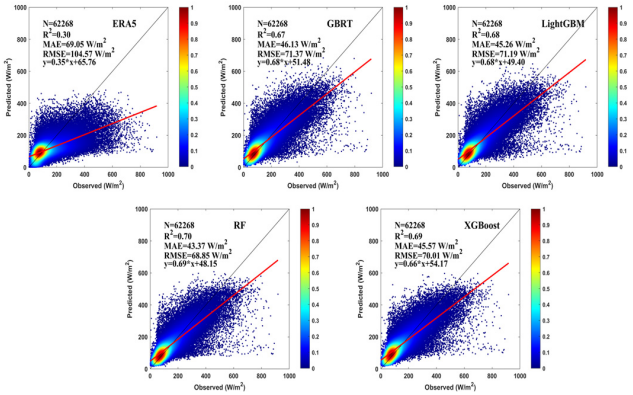


Fig. 7. Comparison with ERA5 product on the independent dataset in the year 2019.

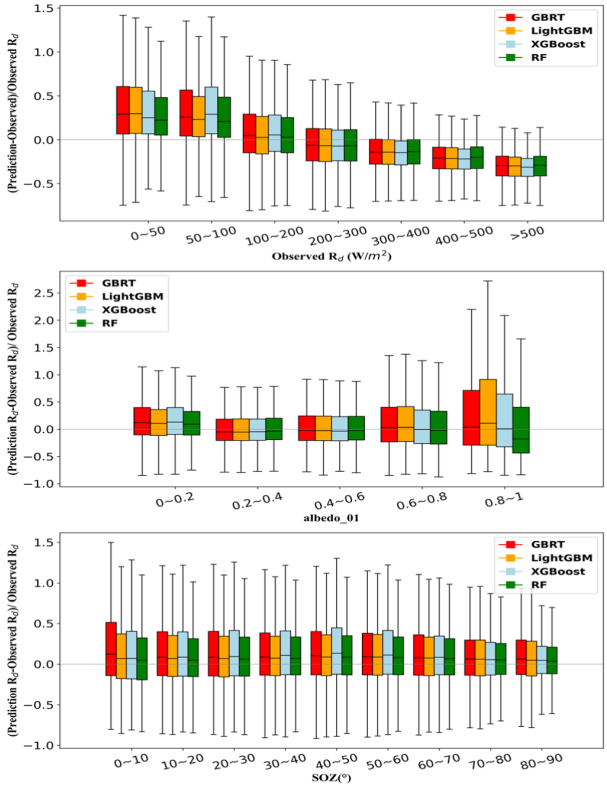


Fig. 8. Model bias varied with the observed R_d, albedo_01, and SOZ.

overestimation at low R_d values while underestimation at high R_d values. At the same time, we saw the change in model error with albedo and SOZ. The model biases were large when the albedo was too small or too large. From radiative transfer theory it is known that albedo is a function of both surface and atmospheric contributions, the surface contribution is generally considered to be constant over short time scales, while the influence of clouds and atmospheric aerosol particles varies considerably [57], [58]. However, there is a small change in the SOZ.

Understanding which factors drive the model predictions can enhance the model's interpretability. The importance of the models is shown in Fig. 9, from Fig. 9 (left), we can see that the albedo_01 and SOZ are the two most important factors for RF, GBRT, and XGBoost. The importance of other features is similar for GBRT and RF, while for XGBoost,

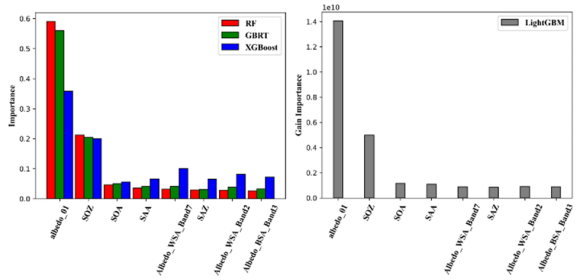


Fig. 9. Importance of input features in four models.

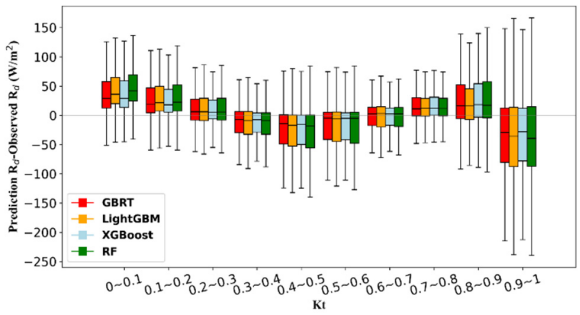


Fig. 10. Model bias depended on Kt.

the features of MCD43A3 have a higher importance, showing that there is a difference in the structure of the three models. The albedo_01 of Himawari-8 is the reflectance of the blue band at the wavelength of 0.47 μm, it contains a lot of information about the cloud and aerosol and is often used for aerosol retrieval [59] and cloud detection [60]. It is, therefore, the largest contributor to the models. LightGBM provides two types of scores for feature importance, they are "Gain" and "Split." The "Gain" quantifies the enhancement in precision introduced by a feature to the branches on which it is located, while the "Split" is the importance determined by the frequency with which a feature is used in a model [53]. From Fig. 9 (right), we can see that the "Gain" importance ranking of features is consistent with the RF and GBRT.

E. Model Bias Under Different Weather Conditions

The clouds in the atmospheric components are considered to be the most important factor affecting the R_s reach at the surface and its partitioning into R_d and direct radiance components [61]. Thus, we evaluated the model under five different meteorological conditions. It turns out that all models did not perform very well in the Clear type but performed best in the Intermediate type, echoing studies showing that R_d from partially cloudy skies is normally substantially higher than the diffuse radiation from clear and typical overcast skies [62]. Furthermore, we built models under clear-sky and not-clear-sky conditions and found that the model under clear-sky cases was greatly improved.

As shown in Fig. 10, we can see the variation of the bias with the Kt values. When Kt is greater than 0.9, which suggests the weather conditions are clear, the model bias is largest because R_d is small under clear skies. The bias is small when Kt is in the categories of 0.3∼0.4, 0.4∼0.5, 0.5∼0.6, and 0.6∼0.7, suggesting that it is usually partly cloudy [63]. This further confirms our results.

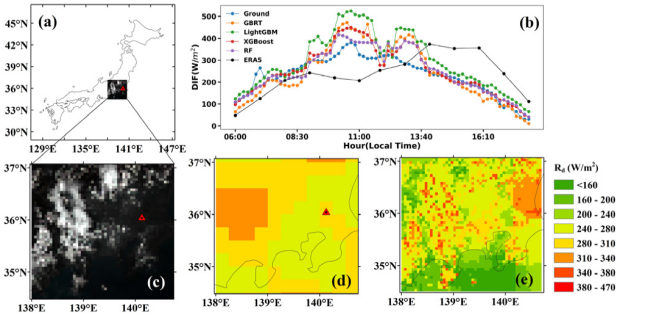


Fig. 11. Prediction of R_d over the target area in the TAT site (red triangle) in Japan on August 1, 2019. (a) Location of the TAT site and target area, (b) diurnal variation, (c) true RGB image at the local time 12:00 on August 1, 2019, and (d) and (e) corresponding spatial distribution of R_d from ERA5 and LightGBM.

F. Model Performance at Different Spatiotemporal Scales and Potential Applications

In this study, we evaluated the model across various time scales, including seasonal and daily scales, and found that all models perform well on different time scales. To further examine the performance of the models on catching the diurnal variations, we selected a TAT site in Japan as the target area, the location is shown in Fig. 11(a). From the temporal scale perspective, the R_d estimated by Himawari-8 can track its dynamical changes within 10 min, compared with ERA5 or other radiation products, our method has more advantages in time scale. For the spatial characteristics, we can see that the estimated R_d from Himawari-8 by LightGBM in Fig. 11(e), which has a resolution of 5 km, could better depict details compared to the ERA5 R_d products, this is consistent with the results in [34]. In summary, results indicate that all four methods reveal the fine spatiotemporal variation of R_d well, in comparison to that of ERA5 with spatial and temporal resolutions of 0.25°h-1.

V. Conclusion

In this study, we adopted four tree-based machine learning methods, including GBRT, LightGBM, RF, and XGBoost to estimate the R_d from Himawari-8 satellite data with a spatiotemporal resolution of 5 km/10 min. First, these methods were evaluated by three different CV methods, it was found that all tree-based methods can predict the diffuse radiation well while the LightGBM has the higher efficiency. We further evaluated the estimates on independent datasets from BSRN and BOM, as well as their performance at different time scales and different weather conditions. We found that all four machine learning methods have good performance for estimating R_d. Second, we compared the results with that of the empirical Starke3 model and DNN model, we found that the tree-based methods have higher accuracy on the independent dataset than the Starke3 model and similar performance with the DNN model. Third, we systemly compared our results with the ERA5 diffuse radiation product, we found our results not only have better performance in accuracy but can also provide higher temporal and spatial resolution. In all, we provide a reliable and simple way to estimate R_d from the Himawari-8 geostationary satellite, which will be of great significance in the applications of R_d.

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